

TANISHK YADAV

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING

Brooklyn, NY

Master of Science in Financial Engineering

05/27

- **Coursework:** Introduction to Derivative Securities, Valuation, Quantitative Methods, Credit Risk and Financial Risk Management, Econometrics and Time Series Analysis, Real Time Risk Management, Algorithmic Trading and High Frequency Finance, Corporate Valuation

SRM UNIVERSITY (100% scholarship)

Amaravati, India

Bachelor of Technology in Computer Science & Engineering (specialization in Artificial Intelligence & Machine Learning)

06/25

GPA: 8.33/10

- **Coursework:** Single/Multi Variable Calculus, Differential Equations, Linear Algebra, Probability, Statistics, Economics

PROGRAMMING / FINANCIAL SKILLS / CERTIFICATIONS

- **Programming Skills:** C, C++, Python (Pandas, NumPy, CCXT), SQL, R, Java, JavaScript, HTML/CSS
- **Machine Learning:** Gaussian Mixture Models (GMM), XGBoost, LSTM, Hierarchical Clustering, TensorFlow, Keras, Scikit-Learn, Random Forest, Time Series Analysis (ARIMA)
- **Financial Concepts:** Regime Switching Models, Tail Risk Hedging, Transaction Cost Analysis, Graph Theory (Arbitrage), Correlation Breakdown, Black-Scholes, Monte Carlo Simulation, VaR, Efficient Frontier
- **Tools and Platforms:** Bloomberg Terminal, Git, VS Code, Jupyter, Excel
- **Certifications:** Bloomberg Market Concepts (BMC), Bloomberg Finance Fundamentals, Akuna Capital Options 101, Financial Markets (HONORS), AI for Investments, IBM Machine Learning

EXPERIENCE

IUDX (INDIA URBAN DATA EXCHANGE), Bengaluru, India

06/23 - 08/23

Summer Intern, Framework Development

- Pioneered development of a secure, cross-platform digital consent collector using SHA-256 asymmetric cryptography, ensuring compliance for critical data aggregation on platform, directly impacting data trustworthiness for ADeX
- Engineered full-stack JavaScript solutions for consent collection and retrieval, ensuring 100% compliance with central government regulations and enabling ADeX to create 38 critical datasets for a government-academia joint venture as a team of 4

RESEARCH / ACADEMIC PROJECTS / FINANCIAL MODELLING

Dynamic Regime-Based Sector Allocation & Tail Risk Hedging | Python, Scikit-Learn, Monte Carlo

- Developed a Regime-Switching Risk Model using Gaussian Mixture Models (GMM) to classify market states into Bull, Transition, and Crisis regimes based on Realized Volatility, VIX, and Yield Spreads
- Demonstrated the failure of static diversification during systemic crises by performing Hierarchical Clustering (Ward's Method) on sector correlation matrices, identifying a "correlation breakdown" toward 1.0
- Backtested a dynamic sector rotation strategy (2018–2026) incorporating transaction costs (10bps); successfully reduced Maximum Drawdown from -35.75% to -19.25% while achieving a Sharpe Ratio of 0.85 (vs. 0.65 Benchmark)
- Validated strategy robustness via 1,000 Monte Carlo simulations (Falsification Testing), statistically proving that the alpha was derived from structural regime avoidance rather than random noise or overfitting

Fx Volatility Forecasting & Arbitrage Identification System | Python, Keras, XGBoost, CCXT

- Engineered a quantitative pipeline to detect Triangular Arbitrage inefficiencies across G10 currency pairs using graph traversal algorithms to identify profitable cyclic paths (e.g., USD→EUR→GBP→USD)
- Benchmarked 7 predictive architectures (ARIMA, LSTM, XGBoost, Random Forest, SVR) on historical OHLC data, utilizing Root Mean Squared Error (RMSE) to select the optimal model for volatility forecasting
- Developed a backtesting engine incorporating transaction costs (basis points) and bid-ask spreads to filter out "ghost" arbitrage opportunities, demonstrating that theoretical alpha is often eroded by execution friction
- Integrated CCXT library to simulate unified API data fetching across multiple exchanges for real-time ticker analysis

POSITIONS OF RESPONSIBILITY

• **Student Council Public Relations Convener**, SRM University, 2022-2023

- Represented 10,000+ students as the primary liaison to the administration, resolving housing and campus grievances and negotiating upgrades to amenities, always prioritizing the well-being of the student community
- Directed cross-functional teams (Hospitality, Sponsorship, PR) to plan and execute major events like annual cultural fests with a \$100,000+ budget, ensuring seamless coordination and stakeholder communication